

Trajectory Framework for Autonomous Quantum Error Correction (QEC)

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In this paper, we present a pedagogical framework to understand autonomous quantum error correction (QEC) via classical phase space dynamics. Using the 3-qubit repetition code, we map dissipative error correction onto gradient flow in Bloch space, using the laws of classical mechanics.

By constructing a Lyapunov potential function, we derive closed-form solutions for error trajectories: correctable single-qubit errors that exhibit straight-line convergence with timescale $\tau = 1/(3\eta)$, uncorrectable multi-qubit logical errors, and phase errors destroying quantum coherence. The analysis reveals a center-of-mass invariance, which protects logical information while individual Bloch vectors suppress errors.

We provide visualization of correction pathways in an effective 2D space and derive the dissipation rate governing error suppression. The techniques we employed like Lyapunov analysis, trajectory classification, are transferable to more sophisticated codes, providing tools for understanding dissipative quantum control across multiple platforms that may offer better design choices.

INTRODUCTION

Classical error-control theory—based on Shannon’s 1948 formulation of communication as a noisy-channel problem[1], continues to provide a conceptual framework upon which modern information theory is built. In any information theory, the error correction mechanism, as well as the protection of information during transmission and storage, is crucial. In classical information theory, error correction is achieved by adding ‘repetition codes’ through an invertible encoding process. This enables us to potentially detect and correct errors in the information. The redundancy can be revealed through ancillary bits. These are the extra bits whose sole purpose is to protect the original codeword and not carry any extra information. To quote an example, a plain repetition code transforms a single bit into three identical bits where the states ‘0’ and ‘1’ become ‘000’ and ‘111’. This transformation imposes a cost in bandwidth and storage but it also provides protection against single-bit-flip errors via ‘majority voting’ during the decoding process.

However, quantum information theory presents a set of fundamentally different challenges. The primary challenge is due to the nature of quantum mechanics itself where the act of measurement collapses the superposed state of the qubit system, making the recovery impossible. Second challenge is the no-cloning theorem[2] which prohibits any quantum operation like this,

$$|\psi\rangle \rightarrow |\psi\rangle \otimes |\psi\rangle. \quad (1)$$

that can take a single state for all states, preventing the straightforward application of repetition codes like in the classical theory. Third, since a qubit occupies a continuum on the Bloch sphere[3], errors on a single qubit form a continuous spectrum depending on the angular coordinates.

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle. \quad (2)$$

On top of all these challenges, a new error type called the phase-flip error is also introduced in the quantum information theory. Due to these fundamental challenges within the theory, physicists have encountered significant difficulties in developing robust quantum error correction codes.

Although quantum error correction is relatively new, its foundation is firmly based on classical error correction. This has made it easier to resolve the aforementioned challenges and development of many robust techniques have been realized over the years.

The first challenge of the measurement problem is addressed by designing interactions between encoded qubits and auxiliary qubits by measuring only the syndrome—information about errors— while leaving the data qubits themselves undisturbed[4][5]. Like in classical error correction, syndromes can be related homomorphically to errors, with syndromes designed to be independent of a quantum state for all states.

The no-cloning restriction is overcome by encoding physical basis states $|0\rangle$ and $|1\rangle$ into larger logical basis states $|0_L\rangle$ and $|1_L\rangle$, as shown by Shor’s nine-qubit code[6]. Also, the mathematical framework of Pauli operators[3] is used to address the continuous nature of quantum errors. Any single-qubit operator can be split into a linear combination of four Pauli matrices: the identity I , the bit-flip operator σ_x , the phase-flip operator σ_z , and their product $\sigma_y = i\sigma_x\sigma_z$. Therefore, in principle, all possible continuous errors can be corrected by correcting these three discrete error types.

Furthermore, a fundamental requirement for QEC is set by the Knill-Laflamme condition[7] which states that a set of eigen-noise corresponding to noise operators N_i and N_j is correctable if and only if it satisfies the condition below for all codewords $|c_p\rangle, |c_q\rangle$ and some complex

numbers λ_{ij} .

$$\langle c_p | N_i^\dagger N_j | c_q \rangle = \lambda_{ij} \delta_{pq}.$$

This guarantees that for different errors, only orthogonal subspaces are mapped to the original code space. Thus, it makes errors uniquely identifiable.

Nonetheless, protecting quantum information against noise addresses only half the challenge. The encoding and decoding procedures themselves can introduce errors, making fault-tolerant quantum computation[8] a necessity. In classical computer systems, fault tolerance is achieved by adding spatial redundancy to reduce failure probability—a concept that applies equally to quantum systems. The accuracy threshold theorem provides the theoretical foundation for scalable quantum computing, stating that if the error rate per qubit falls below a threshold, arbitrarily long quantum computations can be executed with high reliability. This is possible through concatenated codes[9]. Concatenated codes are a hierarchical construction where an $[m,1]$ code encoding one information qubit into m qubits is recursively applied. At L levels of concatenation, an information qubit is encoded into m^L physical qubits. With this, the failure rate decreases as an L -fold exponential function, providing an approach for a scalable quantum computation.

By learning systematic lessons from classical information theory, this paper explores parallels between classical and quantum error correction. This is done by means of classical trajectory interpretation of qubits in Bloch sphere by mapping error correction dynamics with the dynamics of the system. Specifically, we visualize quantum error correction as a phase space flow problem where different error syndromes correspond to distinct trajectories converging toward the codespace attractor as defined by the potential. In the third section, we derive the center-of-mass conservation that protects logical information. This relates to classical systems with exponential relaxation towards equilibrium. The trajectory perspective provides an intuitive understanding of autonomous error correction or any dissipative framework of QEC. In doing so, it reveals structural similarities between classical and quantum information protection that may inform future protocol design.

AUTONOMOUS QEC FORMALISM

The classical optimization and stability analysis techniques from the classical information theory are important tools in quantum information theory. One such example is, reservoir engineering techniques where carefully designed dissipative processes are employed to autonomously drive quantum systems toward desired code states, while continuous weak measurement schemes allow for real-time feedback correction based on classical control algorithms[10]. There are many alternative

paradigms based on this autonomous QEC formalism like dissipation[11]. A recent experiment by Mourik et al.[12] used sideband cooling in trapped ions for autonomous error correction.

Because of discrete qubit architectures, we employ the Bloch sphere representation[13]. In this representation, each qubit corresponds to a point on a three-dimensional sphere, and multi-qubit systems span the product space of these spheres. The codespace and error spaces correspond to orthogonal submanifolds as qubit Bloch vectors. In this geometry, error correction operations become gradient flows that drive the system along unique trajectories from error subspaces toward the codespace manifold, minimizing an appropriately defined Lyapunov function such as potential (V). Specifically, for an n -qubit system, the state is represented by n Bloch vectors $\mathbf{r}_1, \dots, \mathbf{r}_n \in \mathbb{R}^3$, and we construct a potential function $V(\mathbf{r}_1, \dots, \mathbf{r}_n)$ that vanishes on the codespace and increases with distance from it, analogous to classical potential energy.

We demonstrate this framework on the simplest example of a 3-qubit repetition code. Even with its limited error correction (single X-errors only), this 3-qubit repetition code, serves three critical purposes[10]. First, it provides *analytical tractability*: the 9-dimensional Bloch space has closed-form solutions for exact trajectory classification. Second, it enables *methodological demonstration*: techniques developed here generalize to larger codes. Third, it offers *pedagogical clarity*: the geometric structure is visualizable and can be helpful for higher-dimensional cases.

From its analogy to classical wave theory, we can write an equivalent of a continuity equation for the proposed codespace. Let ρ_q define the density of qubit states in a superposed quantum 3-qubit state. Then,

$$\frac{\partial \rho_q}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (3)$$

where $\mathbf{v}$ is the qubit velocity in Codespace region $\mathcal{C}$ given by: $\mathcal{C} = \{\mathbf{r} : \mathbf{r}_1 = \mathbf{r}_2 = \mathbf{r}_3\}$

This is a 3-dimensional subspace in $\mathbb{R}^9$ with Error space regions $\mathcal{E}_i$ given by:

- $\mathcal{E}_1$: qubit 1 flipped ($\mathbf{r}_1 = -\mathbf{r}_2 = -\mathbf{r}_3$)
- $\mathcal{E}_2$: qubit 2 flipped ($\mathbf{r}_2 = -\mathbf{r}_1 = -\mathbf{r}_3$)
- $\mathcal{E}_3$: qubit 3 flipped ($\mathbf{r}_3 = -\mathbf{r}_1 = -\mathbf{r}_2$)

To meet our requirement, we want to define a potential $V(\mathbf{r})$, our Lyapunov function, such that,

1. $V(\mathbf{r}) = 0$ for $\mathbf{r} \in \mathcal{C}$
2. $V(\mathbf{r}) > 0$ for $\mathbf{r} \notin \mathcal{C}$
3. $\frac{dV}{dt} \leq 0$ (always decreasing)

Based on the effective distance between the codespaces (base code and the error code spaces), and the spring constant equivalence set by η , the dissipation rate, a potential appropriate for our geometry could be:

$$V(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3) \propto \frac{1}{2} [(\mathbf{r}_1 - \mathbf{r}_2)^2 + (\mathbf{r}_2 - \mathbf{r}_3)^2 + (\mathbf{r}_3 - \mathbf{r}_1)^2].$$

TRAJECTORY ANALYSIS

To understand the trajectory framework, let's first change to center-of-mass coordinates by defining $\mathbf{R}_{cm}$ as

$$\mathbf{R}_{cm} = \frac{1}{3}(\mathbf{r}_1 + \mathbf{r}_2 + \mathbf{r}_3). \quad (4)$$

From center of mass, the new coordinates are:

$$\mathbf{q}_1 = \mathbf{r}_1 - \mathbf{R}_{cm}, \quad \mathbf{q}_2 = \mathbf{r}_2 - \mathbf{R}_{cm}, \quad \mathbf{q}_3 = \mathbf{r}_3 - \mathbf{R}_{cm}$$

with the constraint condition in cm frame as:

$$\sum_i \mathbf{q}_i = 0$$

According to Breuer et. al [14] the velocity of the flow is given by:

$$\frac{d\mathbf{r}_i}{dt} = -\eta \nabla_{\mathbf{r}_i} V. \quad (5)$$

The sum of all individual velocities is:

$$\sum_{i=1}^3 \frac{d\mathbf{r}_i}{dt} = -\eta \sum_{i=1}^3 \nabla_{\mathbf{r}_i} V = 0.$$

Time derivative of **Eq. 4** together with the above result of sum of all individual velocities, reveals that the center of mass vanishes during the relaxation process. Thus $\mathbf{R}_{cm}$ is the dynamical invariant of the correction process. Thus, for the $\mathbf{q}_i$ coordinate:

$$\frac{d\mathbf{q}_1}{dt} = \frac{d\mathbf{r}_1}{dt} - \frac{d\mathbf{R}_{cm}}{dt} = -\eta(2\mathbf{r}_1 - \mathbf{r}_2 - \mathbf{r}_3).$$

Expressing in terms of $\mathbf{q}_i$ and using the constraint condition gives:

$$\frac{d\mathbf{q}_i}{dt} = -3\eta\mathbf{q}_i.$$

Each relative coordinate decays exponentially with solution given by:

$$\mathbf{q}_i(t) = \mathbf{q}_i(0)e^{-3\eta t}.$$

The exponential part reveals relaxation time towards equilibrium that is characteristic of a dissipative system.

The full solution is: $\mathbf{r}_i(t) = \mathbf{R}_{cm} + \mathbf{q}_i(0)e^{-3\eta t}$

But with center of mass conservation, the relaxation time is given as:

$$\tau = \frac{1}{3\eta}$$

Instead of Newton's second law, we use first-order gradient flow dynamics given the dissipative nature of interaction. This also corresponds to the situation in open quantum systems governed by Lindblad master equations[15], where engineered dissipation and coupling lead to irreversible relaxation toward target states via dissipative stabilization[16].

Also, the rate of change of V , guarantees convergence as suggested by Ticozzi et. al[17]:

$$\frac{dV}{dt} = \sum_{i=1}^3 \nabla_{\mathbf{r}_i} V \cdot \frac{d\mathbf{r}_i}{dt} = -\eta \sum_{i=1}^3 |\nabla_{\mathbf{r}_i} V|^2 < 0.$$

Based on velocity, a Dissipation rate can be defined as:

$$D(t) = \frac{1}{2} \sum_{i=1}^3 \left(\frac{d\mathbf{r}_i}{dt} \right)^2 = \frac{9\eta^2}{2} Q_0 e^{-6\eta t}.$$

where $Q_0 = \sum_{i=1}^3 |\mathbf{q}_i(0)|^2$ defines the initial error magnitude.

ERROR SYNDROME ANALYSIS

The exponential relaxation revealed by the solution above, indicates that all trajectories in the 9-dimensional Bloch space converge toward the codespace manifold $\mathcal{C}$.

For the 3-qubit repetition code, single-qubit bit-flip errors produce distinct correction pathways[3] as can be seen in Figure 1:

Type I: Single-Qubit X Errors (Correctable)

Starting from the logical state $|0_L\rangle = |000\rangle$ with Bloch vectors $\mathbf{r}_1 = \mathbf{r}_2 = \mathbf{r}_3 = (0, 0, 1)$, the three correctable errors are:

- **Error $\mathcal{E}_1$:** $X_1|000\rangle = |100\rangle$

$$\mathbf{r}_1(0) = (0, 0, -1), \quad \mathbf{r}_2(0) = \mathbf{r}_3(0) = (0, 0, 1)$$

- **Error $\mathcal{E}_2$:** $X_2|000\rangle = |010\rangle$

$$\mathbf{r}_1(0) = \mathbf{r}_3(0) = (0, 0, 1), \quad \mathbf{r}_2(0) = (0, 0, -1)$$

- **Error $\mathcal{E}_3$:** $X_3|000\rangle = |001\rangle$

$$\mathbf{r}_1(0) = \mathbf{r}_2(0) = (0, 0, 1), \quad \mathbf{r}_3(0) = (0, 0, -1)$$

$$\text{with } \mathbf{R}_{cm} = \frac{1}{3}(0, 0, 1) \text{ for all three.}$$

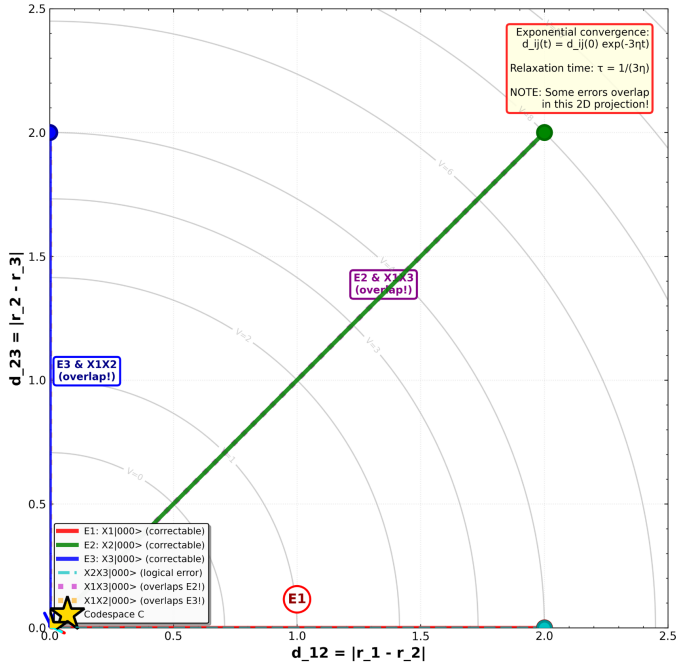


FIG. 1. 2D representation of phase space trajectories with equipotential contours of $V(r_1, r_2, r_3)$ projected onto the (d_{12}, d_{23}) plane where $d_{12} = |r_1 - r_2|$ and $d_{23} = |r_2 - r_3|$. The green trajectory illustrates convergence from the correctable error subspaces (E1: $X_1|000\rangle$, E2: $X_2|000\rangle$, E3: $X_3|000\rangle$) toward the codespace $\mathcal{C}$ (marked with a star). Overlapping errors (E2 & X_1X_3 , E3 & X_1X_2) and logical errors (X_2X_3) are also indicated.

These error states satisfy the Knill-Laflamme orthogonality condition in the Hilbert space $(\mathbb{C}^2)^{\otimes 3}$:

$$\langle 100|010\rangle = \langle 010|001\rangle = \langle 001|100\rangle = 0$$

All three trajectories exhibit exponential convergence to the same center-of-mass $\mathbf{R}_{cm} = \frac{1}{3}(0, 0, 1)$, which represents the original logical state.

Type II: Multi-Qubit Errors (Uncorrectable)

Two-qubit errors exceed the code distance $d = 3$ and lead to logical errors[18]. For example, $X_1X_2|000\rangle = |110\rangle$ gives:

$$\mathbf{r}_1(0) = \mathbf{r}_2(0) = (0, 0, -1), \quad \mathbf{r}_3(0) = (0, 0, 1)$$

with center-of-mass: $\mathbf{R}_{cm} = -\frac{1}{3}(0, 0, 1)$

The system is driven to this center-of-mass, which is closer to $|1_L\rangle = |111\rangle$ than to the original $|0_L\rangle = |000\rangle$. This represents a *logical error* because the majority voting has failed. The gradient flow minimizes the potential V , but it converges to the wrong codeword.

Type III: Phase Errors (Uncorrectable)

The 3-qubit bit-flip code does not protect against Z errors (phase flips). A phase error $Z_1(\alpha|000\rangle + \beta|111\rangle) = \alpha|000\rangle - \beta|111\rangle$ creates transverse deviations in the Bloch

representation (components in the xy -plane). Consequently, the system converges to a state with destroyed superposition.

CONCLUSION AND FUTURE SCOPE

The trajectory framework is a powerful visualization tool. In conservative classical systems (e.g., harmonic oscillators), trajectories are closed orbits with conserved energy. In contrast, in autonomous QEC, all trajectories spiral inward toward the codespace attractor, with the potential $V(t)$ decaying exponentially.

The conservation of center-of-mass $\frac{d\mathbf{R}_{cm}}{dt} = 0$ represents the analogue of classical conservation law. This conserved quantity encodes the logical qubit state—the very information the code is designed to protect. While individual Bloch vectors evolve dynamically to suppress errors, their average (the logical state) remains invariant, as they should.

The characteristic relaxation timescale constrains how rapidly errors are suppressed. For effective error correction, we require:

$$\tau_{corr} = \frac{1}{3\eta} \ll \tau_{error}$$

where τ_{error} is the typical time between error events. This requires strong dissipation (η large). The strong coupling between qubits and a dissipative environment can be created in an experimentally designed neutral atom or trapped ion platform[10]. For neutral atoms, a combination of Rydberg-mediated interactions and optical pumping can provide dissipative stabilization[11]. Recent below-threshold demonstrations with superconducting qubits have identified rare correlated error bursts occurring approximately once per hour ($\sim 3 \times 10^9$ cycles). These localized events, affecting neighborhoods of ~ 30 qubits with decay timescales of 400–700 μs , arise from mechanisms not yet fully understood[19].

While 3-qubit repetition error correction model is limited to single X-error correction, this framework can extend to experiments including biased-noise architectures where autonomous X-suppression reduces measurement overhead [20], cat qubits and bosonic codes with continuous-variable dissipation [21], and surface codes with engineered multi-qubit stabilizer dissipation [22]. The characteristic timescale τ and dissipation rate $D(t)$ can be generalized to these larger codes for quantitative predictions or for efficient error correction design. It might offer architectural advantages in specific fields of quantum information processing.

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